MD. NADIM HOSSAIN

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ACADEMIC CREDENTIAL

Rajshahi University of Engineering and Technology (RUET) *March 2022 - August 2026*
Bachelor of Science (BSc.) in Materials Science & Engineering
GPA: 3.16/4.00
Recipient of University Merit Scholarship (2×)

Saint Joseph Higher Secondary School, Dhaka *2017 – 2019*
Higher Secondary School Certificate (HSC)
GPA: 5.00/5.00

RESEARCH INTEREST

- Lithium-ion Batteries PEM Fuel Cells Hydrogen Production and Storage Solar Energy Conversion Nanomaterials Electrochemistry MXene-based Energy Materials

RESEARCH EXPERIENCE

Department of Materials Science Engineering, RUET
Undergraduate Thesis and Research Project *Sept. 2024 - Aug. 2026*

- Fabrication and Investigation of Nitrogen-Phosphorus Co-Doped SnO_2 Anodes for Enhanced Electrochemical Performance and Cycling Stability in Lithium-Ion Batteries.
- Investigation of doping-induced defect chemistry, structural modification, and electrochemical behavior of SnO_2 anode materials from atomic-scale interaction to practical cell performance.

PUBLICATIONS AND MANUSCRIPTS

Under Review / Submitted

- Hossain, M. N.; Hossen, M. M. *Between-Laboratory Variation Can Exceed Dopant Effects: A Validation-Aware Meta-Analysis of Literature-Derived ZnO Photocatalysts for Hydrogen Production.* Submitted to Digital Discovery.
- Hossain, M. N.; Hossen, M. M.; Siddika, S. *Study-Aware Small-Data Machine Learning for Initial Coulombic Efficiency Prediction of Doped SnO_2 Lithium-Ion Battery Anodes.* Submitted to 7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), IEEE, 2027.
- *Nanomaterials for Advanced Batteries and Supercapacitors: Performance Enhancement, Scalability Challenges, and AI-Assisted Material Optimization.* Submitted to 7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), IEEE, 2027.

In Preparation

- Metal Oxide as an Anode Catalyst Support for Proton Exchange Membrane (PEM) Fuel Cells.
- Recent Trends of MXene-Based Materials and Derivatives for Sustainable Next-Generation Energy Technologies: Energy Storage, Absorption, Conversion, and Harnessing.

PROJECTS

Case Study on Doping in Lithium-Ion Battery Anodes: Defect Chemistry, Coupled Architectures, and Evidence at Cell Scale

SKILL SUMMARY

Programming & Data Analysis	:	Python, Machine Learning, MATLAB, Matplotlib, X'Pert HighScore Plus, ImageJ, OriginPro
Computational Materials Tools	:	Materials Studio, VESTA
Materials Analysis & Characterization	:	XRD, FESEM, FTIR, UV–Vis Spectroscopy, GC-MS, ICP-OES, DLS, Particle Size Analyzer, DTA, TGA data analysis
Electrochemical Analysis	:	CV, GCD, EIS, Nyquist plot analysis, R_s/R_{ct} evaluation, charge–discharge behavior
CAD Tools	:	SolidWorks, Autodesk AutoCAD
Documentation & Collaboration	:	Microsoft Office Suite, Canva, Mendeley Citation Manager
Leadership	:	Negotiation, Crisis Mitigation, Project Management, Budget Management, Event Management
Soft Skills	:	Technical Report Writing, Public Speaking, Storytelling, Critical Thinking

RELEVANT COURSEWORK

- Crystallography and Structure of Materials
- Crystal Defects, Deformation, and Fracture
- Glass and Ceramic Engineering
- Ceramics and Advanced Applications
- Surface Engineering of Materials
- Materials for Energy Generation, Conversion, and Storage
- Materials Characterization
- Physical Properties of Materials
- Iron and Steel Making
- Foundry Engineering
- Corrosion and Degradation of Materials
- Physical Metallurgy and Heat Treatment
- Metallic Alloys and Material Selection
- Industrial Metal Working Process
- Polymer and Composite Engineering
- Heat and Mass Transfer
- Fluid Mechanics and Hydraulic Machineries
- Electrical Machines and Electronics
- Advanced Materials and Nanotechnology

INDUSTRY EXPERIENCE

Shahriar Steel Mills Ltd.
Intern *March 2025*
Konapara, Jatrabari, Dhaka-1362, Bangladesh

BCSIR Laboratories, Rajshahi, Bangladesh
Training on: FESEM, GC-MS, ICP-OES, FTIR, and UV-Vis-NIR Spectroscopy Techniques.

REFERENCES

1. Dr. Md. Aminul Islam
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